

Why does the spectral norm equal the largest singular value?

Asked 11 years, 1 month ago Modified 2 years, 8 months ago Viewed 77k times



This may be a trivial question yet I was unable to find an answer:

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$$\|A\|_2 = \sqrt{\lambda_{\max}(A^*A)} = \sigma_{\max}(A)$$



where the spectral norm $\|A\|_2$ of a complex matrix A is defined as

$$\max \{\|Ax\|_2 : \|x\| = 1\}$$



How does one prove the first and the second equality?

[linear-algebra](#) [matrices](#) [svd](#) [singular-values](#) [spectral-norm](#)

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edited Aug 21, 2018 at 11:05
Rodrigo de Azevedo
1

asked Nov 30, 2013 at 10:43
mathemage
850 1 7 15

What are your thoughts on the second equality? – Git Gud Nov 30, 2013 at 11:35

@GitGud Oh, does that mean singular values are defined in this way, i.e., the second equality is a definition? – mathemage Nov 30, 2013 at 12:23

It's not a definition. It's a direct consequence of the definition. Can you see this? – Git Gud Nov 30, 2013 at 12:26

math.stackexchange.com/questions/482170/2-norm-vs-operator-norm – Prahlad Vaidyanathan Nov 30, 2013 at 17:11

@GitGud One normally can't if one does not know about the Courant-Fischer's characterisation. – Algebraic Pavel Dec 1, 2013 at 2:37

2 Answers

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First of all,

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$$\sup_{\|x\|_2=1} \|Ax\|_2 = \sup_{\|x\|_2=1} \|U\Sigma V^T x\|_2 = \sup_{\|x\|_2=1} \|\Sigma V^T x\|_2$$



since U is unitary, that is, $\|Ux_0\|_2^2 = x_0^T U^T U x_0 = x_0^T x_0 = \|x_0\|_2^2$, for some vector x_0 .



Then let $y = V^T x$. By the same argument above, $\|y\|_2 = \|V^T x\|_2 = \|x\|_2 = 1$ since V is unitary.

$$\sup_{\|x\|_2=1} \|\Sigma V^T x\|_2 = \sup_{\|y\|_2=1} \|\Sigma y\|_2$$

Since $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$, where σ_1 is the largest singular value. The max for the above, σ_1 , is attained when $y = (1, \dots, 0)^T$. You can find the max by, for example, solving the above using a Lagrange Multiplier.

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edited Mar 31, 2019 at 7:02
Martin Sleziak
55.2k 20 203 381

answered Jan 31, 2018 at 22:44
yulunz
601 5 2



Put $B = A^*A$ which is a Hermitian matrix. A linear transformation of the Euclidean vector space E is Hermite iff there exists an orthonormal basis of E consisting of all the eigenvectors of B . Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of B and $\{e_1, \dots, e_n\}$ be an orthonormal basis of E . Denote by λ_{j_0} to be the largest eigenvalue of B .

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For $x = a_1 e_1 + \dots + a_n e_n$, we have $\|x\| = \langle \sum_{i=1}^n a_i e_i, \sum_{i=1}^n a_i e_i \rangle^{1/2} = \sqrt{\sum_{i=1}^n a_i^2}$ and $Bx = B(\sum_{i=1}^n a_i e_i) = \sum_{i=1}^n a_i B(e_i) = \sum_{i=1}^n \lambda_i a_i e_i$. Therefore:



$$\begin{aligned} \|Ax\| &= \sqrt{\langle Ax, Ax \rangle} = \sqrt{\langle x, A^* A x \rangle} = \sqrt{\langle x, Bx \rangle} = \sqrt{\langle \sum_{i=1}^n a_i e_i, \sum_{i=1}^n \lambda_i a_i e_i \rangle} \\ &= \sqrt{\sum_{i=1}^n a_i \lambda_i a_i} \leq \max_{1 \leq j \leq n} \sqrt{\lambda_j} \times (\|x\|) \end{aligned}$$

So, if $\|A\| = \max \{\|Ax\| : \|x\| = 1\}$ then $\|A\| \leq \max_{1 \leq j \leq n} \sqrt{\lambda_j}$. (1)

Consider: $x_0 = e_{j_0} \Rightarrow \|x_0\| = 1$ so that $\|A\|^2 \geq \langle x_0, Bx_0 \rangle = \langle e_{j_0}, B(e_{j_0}) \rangle = \langle e_{j_0}, \lambda_{j_0} e_{j_0} \rangle = \lambda_{j_0}$. (2)

Combining (1) and (2) gives us $\|A\| = \max_{1 \leq j \leq n} \sqrt{\lambda_j}$ where λ_j is the eigenvalue of $B = A^*A$

Conclusion:

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^*A)} = \sigma_{\max}(A)$$

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edited Apr 21, 2022 at 9:24
user1868607
6,085 4 34 66

answered Nov 30, 2013 at 14:09
An Khuong Doan
2,145 15 12

Why is A^*A a unitary matrix? For A as a zero matrix or a general real diagonal matrix, this doesn't have to be unitary, right? – mathemage Nov 30, 2013 at 16:29

2 That's my mistake. That should be Hermite. Got fixed! – An Khuong Doan Nov 30, 2013 at 16:58 ↗

2 And (2) is still blur to me: how can we say

$$\|A\| \geq \langle x, Bx \rangle = \langle e_{j_0}, B(e_{j_0}) \rangle$$

– sleeve chen Jan 21, 2015 at 0:01 ↗

1 @sleevechen for (1) we can say it because $\|A\|$ is the maximum of all the elements in the set, and it was just shown that each element is $\leq$ the largest singular value times $\|x\| = 1$. For (2), since $\|A\|$ is the maximum, it must be $\geq$ than any particular element, namely $\|Ae_{j_0}\| = \|Ae_{j_0}\| = \langle Ae_{j_0}, Ae_{j_0} \rangle = \langle e_{j_0}, A^*Ae_{j_0} \rangle = \langle e_{j_0}, B(e_{j_0}) \rangle$ – ignoramus Jun 8, 2016 at 11:40

1 @AnKhuongDoan, Could you explain why on page 16 in math.ucla.edu/~wotaovin/summer2016/4_proximal.pdf it says: "note: the spectral norm returns the square root of the max eigenvalue of $A^T A$, may not equal the max singular value for asymmetric matrices."? – Royi Sep 25, 2017 at 8:17